ReGate: Mitigating Hallucinations in Visual Grounding via Metamorphic Counterfactual Supervision

Anonymous submission

Abstract

Visual grounding models are typically trained under the assumption that the queried referent exists, causing unsupported localization when a query conflicts with the image or the target is unobservable. To address this problem, we propose ReGate: *reason-attributed evidence gating*, a two-stage framework for selective grounding that either returns a supported label and box or abstains with an explicit cause: visual contradiction or target unavailability. At its core, ReGate employs *metamorphic counterfactual supervision* (MCS), which turns seven deterministic relations into supervision by specifying how evidence states and spatial targets should remain invariant or change under controlled transformations. This construction yields SelectGround-Bench with 7,072 training instances, construction-defined relation tests, and same-source counterfactual groups with contrasting outcomes under shared provenance. ReGate first adapts Qwen3-VL to the resulting three-state grounding objective and then fits a compact pre-decoding gate for explicit routing. Across six training-disjoint probes, ReGate delivers strong selective grounding and cross-domain transfer: a development-selected operating point reduces the SFT-native false-commitment rate by 52.0% while increasing supported retention, and matched retraining raises metamorphic satisfaction from 0.527 to 0.820. On the comprehensive test, under the same instruction and output schema, ReGate also outperforms GPT-5.6 Sol in false commitment (0.132 versus 0.302), Strict-3 accuracy, and end-to-end grounding. These results establish metamorphic counterfactual supervision as an effective and reusable supervision paradigm for evidence-aware VLM adaptation.

1 Introduction

Vision–language models (VLMs) can accurately localize natural-language expressions (Radford et al. 2021; Li et al. 2023a; Liu et al. 2023; Wang et al. 2024; Liu et al. 2024), but standard referring-expression benchmarks usually assume that the requested target exists and is visible (Mao et al. 2016). When a query conflicts with the image, refers to an absent target, or lies outside the observable region, a model may still emit a plausible label and box: a spatially auditable but unsupported commitment that we term *unsupported grounding hallucination*. Unlike mislocalizing a supported target, the correct action is to withhold localization.

To address this problem, we propose ReGate: *reason-attributed evidence gating*, for *reason-attributed selective grounding*. ReGate predicts *supported grounding*, *visual contradiction*, or *target unavailable*; supported cases require a correct label and region, whereas unsupported cases require abstention with the corresponding cause. This three-state formulation extends binary answerability by determining both whether grounding is warranted and why it should be withheld, complementing strong localization models with an explicit evidence-state decision before localization.

A central challenge is to learn evidence distinctions rather than source identity. Independently collected negatives may expose scene, acquisition, or provenance shortcuts. This issue is especially demanding in agriculture, where fine-grained symptoms, small objects, role ambiguity, and target absence complicate the boundary between localization and abstention. We therefore introduce *metamorphic counterfactual supervision* (MCS). Seven deterministic relations specify how wording, appearance, geometry, query compatibility, target visibility, and spatial annotations should remain invariant or change under controlled transformations (Chen, Cheung, and Yiu 1998; Segura et al. 2016). Their instantiation, SelectGround-Bench, produces annotation-defined supervision and relation tests; same-source counterfactual groups assign contrasting outcomes under shared provenance, directly supervising query–evidence compatibility. By reusing target-identity and localization annotations, MCS also offers a scalable auxiliary supervision task for VLM pretraining and adaptation.

Built on Qwen3-VL-8B-Instruct, ReGate first adapts the VLM for structured grounding and three-state prediction, then freezes the adapted backbone and fits a compact pre-decoding gate. The gate routes supported inputs to the shared grounding decoder and returns an explicit abstention reason otherwise, combining representation-level hallucination mitigation with high-retention routing.

Our contributions are threefold:

- We formulate unsupported grounding hallucination as an annotation-verifiable three-state problem unifying localization, contradiction detection, and target unavailability.
- We introduce MCS and SelectGround-Bench, deriving reusable evidence-state supervision and relation

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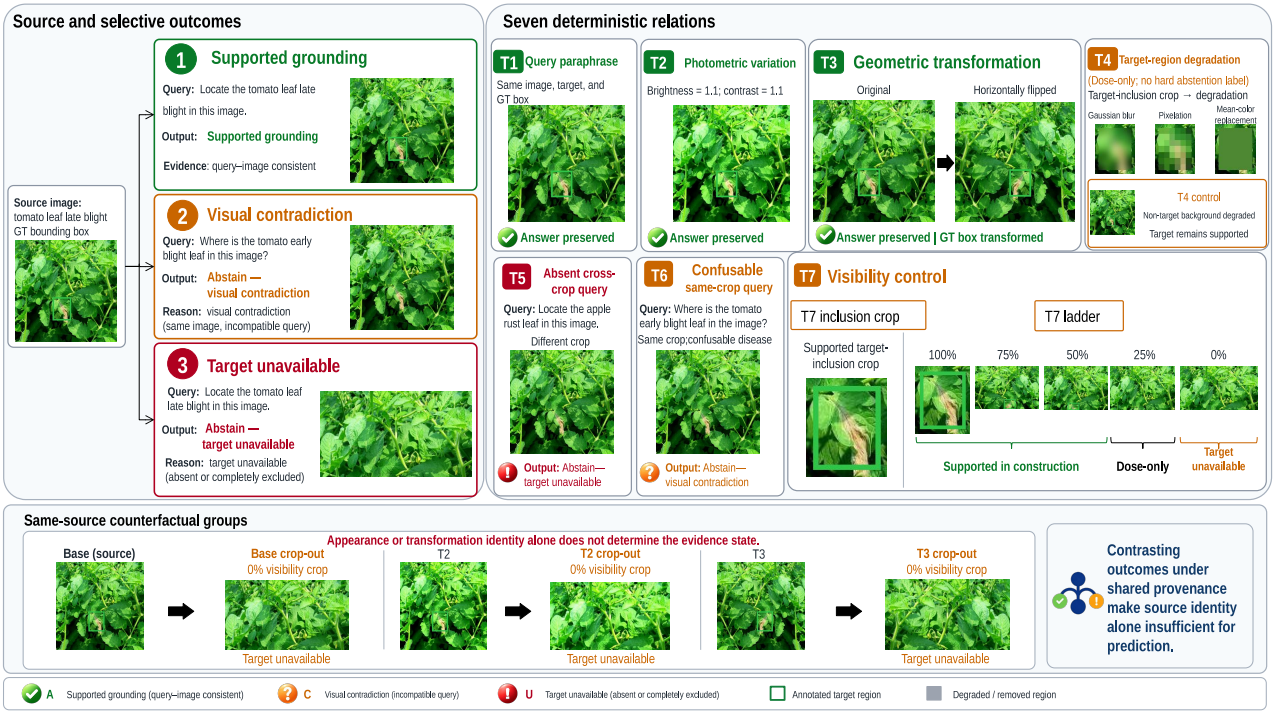


Figure 1: Task and benchmark construction. Same-source transformations yield supported grounding, visual contradiction, or target unavailability while controlling query compatibility and target visibility.

tests from seven deterministic transformations and same-source counterfactual groups.

- We develop the two-stage ReGate framework, which substantially reduces unsupported commitment while improving label-gated localization and supported retention.

Experiments span a comprehensive in-domain test, controlled probes, source-disjoint PlantWild, and image-disjoint RefCOCOg, comparing the frozen backbone, native post-adaptation decoding, and explicit routing; grouping, representation, evidence-removal, and visibility studies further isolate the gains. A development-selected operating point reduces SFT-native false commitment by 52.0% while increasing retention, and matched retraining raises metamorphic satisfaction from 0.527 to 0.820. Under the same instruction and schema, ReGate also outperforms GPT-5.6 Sol in false commitment (0.132 versus 0.302), Strict-3, and end-to-end grounding.

2 Related Work

Multimodal hallucination and unsupported visual queries. VLM hallucination includes objects, attributes, and relations that are unsupported by the visual input (Li et al. 2023b; Bai et al. 2024). POPE probes object hallucination through existence-oriented questions (Li et al. 2023b), while Antidote targets counterfactual-presupposition and object-perception hallucinations through synthetic post-training (Wu et al. 2025). Grounding has also been investigated as a hallucination remedy, yet stronger grounding does not automatically reduce hallucination in open-ended generation (Geigle, Timofte, and Glavaš 2024). We focus on a spatially auditable failure: committing to a label and box when benchmark-defined evidence is contradictory or unavailable.

Selective prediction and visual-evidence verification. Selective prediction controls the trade-off between coverage and predictive risk (Geifman and El-Yaniv 2017, 2019). Reliable VQA extends this principle with learned selectors and answer-verification mechanisms (Whitehead et al. 2022; Dancette et al. 2023; Khan and Fu 2024; Srinivasan et al. 2024). SIEVES explicitly scores localized visual evidence for selective prediction across out-of-distribution tasks (Rodriguez and Rohrbach 2026). These methods mainly estimate whether an answer should be trusted. ReGate advances selective prediction into grounding by predicting one of three evidence states before localization and distinguishing annotation-defined visual contradiction from unavailable target evidence.

Generalized and counterfactual grounding. Conventional referring-expression comprehension assumes that a valid target is present. GREC relaxes this assumption to allow zero, one, or multiple targets, and RECANTFormer addresses the resulting variable-target prediction problem (He et al. 2023; Hemanthage et al. 2024). Counterfactual REC constructs queries whose referents cannot be found and learns a counterfactual label alongside localization (Yu and Li 2024). Together, these works establish that grounding should not force a region when the query lacks a valid referent. Our formulation extends this direction with first-class separation of visual contradiction and target unavailability,

reason-attributed abstention, and strict evaluation under spatially transformed annotations.

Counterfactual and metamorphic supervision. Counterfactual vision–language training uses minimally changed image–text pairs to improve fine-grained discrimination and compositional reasoning (Zhang et al. 2025; Wu et al. 2025). Metamorphic testing specifies expected output relations under controlled input transformations when a complete direct oracle is unavailable (Chen, Cheung, and Yiu 1998; Segura et al. 2016). Relational validation across cyber–cyber digital twins further shows how controlled system variants can verify required properties under counterfactual behavior (Ahlgren et al. 2021). SELECTGROUND-BENCH combines these principles through deterministic, annotation-defined transformations that intervene on query compatibility, target visibility, and evidence preservation while retaining source provenance. The resulting records support both sample-wise supervision and relation-stratified strict testing without generative counterfactual imagery.

Agricultural visual diagnosis and grounding. Plant-disease recognition varies substantially in illumination, background, scale, and acquisition source (Mohanty, Hughes, and Salathé 2016; Singh et al. 2020; Wei et al. 2024), while agricultural VLM and grounding resources increasingly evaluate multimodal reasoning and localization (Shinoda et al. 2025; Yang et al. 2025; Awais et al. 2025; Haghighat, Saleh, and Azghadi 2026; Li et al. 2026). These conditions make agriculture a natural stress test for all three evidence states and source shift, while COCO-derived training and RefCOCOg evaluation test transfer to general visual categories.

3 SELECTGROUND-BENCH

3.1 Task and Metamorphic Construction

SELECTGROUND-BENCH operationalizes metamorphic counterfactual learning by deriving evidence-state supervision from datasets with target-identity and location annotations. Given image I and query q, the reference outcome is $y^* \in \{\textsc{Answer}(l^*, b^*), \textsc{Abstain}(\rho^*)\}$, where l^* is the target label, $b^* \in [0, 1000]^4$, and ρ^* is visual contradiction or target unavailability. A prediction o yields a decision $d(o)$ and, when applicable, a label $l(o)$, box $b(o)$, or reason $\rho(o)$. Contradiction is assigned only when source annotations identify a configured confusable class while excluding the queried class. Construction-verified absence, an incompatible cross-crop query, or complete target removal yields unavailability. Thus, once the admissible label space is fixed, source annotations and deterministic transformations define every reference outcome independently of the evaluated model.

A fixed symmetric canonicalizer evaluates label equivalence, and invalid outputs receive no semantic, spatial, or reason credit. With $\mathsf{ans} = \textsc{Answer}$, $\mathsf{abs} = \textsc{Abstain}$, and $d = d(o)$, $l = l(o)$, $b = b(o)$, and $\rho = \rho(o)$, define

$$C_S = \mathbf{1}[d = \mathsf{ans}, l = l^*], C_R = \mathbf{1}[d = \mathsf{abs}, \rho = \rho^*], \quad C_G = C_S\, \mathbf{1}[\mathrm{IoU}(b, b^*) \geq 0.5]. \tag{1}$$

Figure 1 summarizes the seven deterministic relations. T1–T3 preserve the target under paraphrasing, photometric variation, and geometric transformation. T4 varies target degradation, with a supported non-target degradation control. T5 pairs an observed crop with an absent cross-crop query; T6 substitutes a confusable same-crop class; and T7 controls retained target visibility. Base, photometric, and geometric appearances are additionally paired with complete crop-outs, preventing appearance or transformation identity from determining the state. Hard labels are used only when annotations and transformations determine the semantic and, when applicable, spatial outcome; other degradation and partial-visibility records are reserved for graded-response analysis.

T6 leaves the image unchanged and admits a substituted query only when it belongs to the fixed same-crop confusable set, is absent from all source-image annotations, and remains in the retained label space. T7 requests $100\%, 75\%, 50\%, 25\%$, and 0% retained visibility and remaps the box into derived-image coordinates. The 100%, 75%, and nominal 50% records achieving at least 50% visibility are supported; complete exclusion is unavailable; the remaining intermediate records are graded only. A separate T7 inclusion crop retains the target with approximately 10% surrounding context.

3.2 Same-Source Data and Evaluation Protocol

The principal contradiction and visibility records are grouped by source. T6 combines different queries with the same image, whereas T7 combines target-retaining and target-excluding views of the same annotation. Hence $\mathrm{src}(I_a) = \mathrm{src}(I_u)$ but $y_a \neq y_u$, making source identity insufficient within a group. The grouped subset contains $1{,}907$ instances: 372 supported anchors, 973 contradiction variants, and 281 target-retaining plus 281 target-excluding variants. Some anchors participate in both T6 and T7; the remaining records supply additional annotation-derived supervision.

Before resampling, the assembled training split contains $6{,}545$ instances and $6{,}138$ distinct records. Training-only resampling expands the 973 contradiction instances to $1{,}500$, adding 527 instances and yielding the $7{,}072$-instance composition in Table 1; COCO-derived records use canonical categories, instances, and boxes (Lin et al. 2014). The agricultural label space is fixed before splitting using the frozen base VLM on the verified PlantDoc pool: 20 of 24 observed categories meet the predefined category-level threshold of 0.40 correct label-and-localization accuracy at IoU 0.5. Categories are retained or excluded in full, without record-level filtering or downstream results, and the resulting universe is then split by source image. All derivatives of a source remain in one partition; dose-only records remain separate; and resampling follows group assignment.

Table 2 summarizes six training-disjoint probes. The comprehensive probe evaluates all three states and eleven non-base relation variants. L1 isolates same-crop contradiction, and L2 pairs retained evidence with complete exclusion. L3 contains 750 visibility-controlled records: fixed-point metrics use 183 fully visible and 148 fully excluded endpoints, while the remaining 419 define the graded response. PlantWild (Wei et al. 2024) uses its official field-image test split and retains the 12 disease classes represented during SFT, producing 144 supported, 144 unavailable, and 120 contradictory records; its image-level annotations support selective-state and reason evaluation but not box scoring. Re-

Component	Count	Dec.	State
Contradiction counterfactuals	1,500	Abs.	Contr.
Complete target exclusions	669	Abs.	Unavail.
Cross-crop unavailable queries	1,500	Abs.	Unavail.
Appearance-matched crop-outs	200	Abs.	Unavail.
COCO-derived unavailable	500	Abs.	Unavail.
COCO-derived supported	1,100	Ans.	Supp.
Visibility-controlled supported	300	Ans.	Supp.
Original/transformed supported	1,303	Ans.	Supp.
Total	7,072	38:62	2,703/1,500/2,869

Table 1: Effective training composition after training-only resampling. In the total row, Dec. is Answer:Abstain and State is Supported/Contradiction/Unavailable.

Probe	S/U	Evaluation focus
L1	205/205	Same-crop discrimination (T6)
L2	395/395	Evidence retention/exclusion
L3	183/148	$100\%/0\%$ endpoints; partial-level curve
Comp.	1,323/1,323	Comprehensive in-domain test
PW	144/264	PlantWild source-disjoint field test
Ref.	181/181	RefCOCOg image-disjoint grounding

Table 2: Training-disjoint probes (S/U: supported/unsupported).

fCOCOg (Mao et al. 2016) contributes 181 original expressions, categories, and boxes, paired with 181 image-disjoint COCO images whose annotations exclude the queried category. COCO training images are disjoint from RefCOCOg evaluation images, and PlantWild is screened for exact and perceptual overlap.

3.3 Metrics

Let $\mathcal{D}_{\text{sup}}$ and $\mathcal{D}_{\text{unsup}}$ denote the fixed supported and unsupported reference universes. The false-commitment rate (FCA) and supported retention Ret_S are the answer-selection rates on $\mathcal{D}_{\text{unsup}}$ and $\mathcal{D}_{\text{sup}}$, respectively; $\text{A_FU} = 1 - \text{Ret}_S$. Selective MCC is the standard Matthews coefficient on the binary answer/non-answer decision. The two task-specific aggregate metrics are

$$\text{E2E-mIoU} = |\mathcal{D}_{\text{sup}}|^{-1} \sum_{i \in \mathcal{D}_{\text{sup}}} C_S(o_i)\, \text{IoU}_0(b(o_i), b_i^*) \quad (2)$$

$$\text{Strict-3} = \frac{\sum_{i \in \mathcal{D}_{\text{sup}}} C_S(o_i) + \sum_{i \in \mathcal{D}_{\text{unsup}}} C_R(o_i)}{|\mathcal{D}_{\text{sup}}| + |\mathcal{D}_{\text{unsup}}|} \quad (3)$$

where IoU_0 maps invalid boxes to zero. E2E-mIoU therefore jointly measures answering, label correctness, and localization, whereas Strict-3 measures three-state semantic correctness without localization.

For probe p, let $\mathcal{V}_p$ be its instantiated relation variants, $\mathcal{D}_{p,v}$ their records, and $m_v(o_i)$ strict compliance with the construction-defined semantic and spatial outcome. Macro metamorphic test satisfaction is

$$\text{MTS}_p = \frac{1}{|\mathcal{V}_p|} \sum_{v \in \mathcal{V}_p} \frac{1}{|\mathcal{D}_{p,v}|} \sum_{i \in \mathcal{D}_{p,v}} m_v(o_i). \quad (4)$$

MTS weights each variant equally and does not credit predictions that are mutually consistent but jointly incorrect. Supported relations require C_G; complete crop-outs require unavailability; T5 and T6 require their designated reasons; and supported T7 records additionally require the spatial criterion. L3 separately measures response across all available visibility levels.

For generative interfaces, a final invalid parse is a non-answer for selection metrics and receives no semantic, reason, or spatial credit. For ReGate, the gate route defines binary selection: a malformed or abstention-form candidate on an answer route remains an answer selection but receives zero correctness credit. Supported answer-and-grounding accuracy and unsupported-reason accuracy are reported only as secondary decompositions in the construction contrast.

4 Reason-Attributed Evidence Gating

MCS is model-independent at the level of construction operators and reference outcomes. ReGate is a backbone-compatible two-stage framework for VLMs exposing a query-conditioned hidden representation and a structured grounding decoder; our implementation uses Qwen3-VL-8B-Instruct (Bai et al. 2025). Stage 1 uses metamorphic evidence-state supervision to jointly adapt the VLM representation and an auxiliary state gate. Stage 2 freezes the adapted VLM, validates the readout position, and separately fits the inference gate on the resulting stationary representation. Supported inputs proceed to grounding generation, whereas unsupported inputs return the attributed cause without decoding. Figure 2 summarizes the framework.

4.1 Metamorphic Evidence-State Adaptation

For an image–query pair, the readout is the hidden state of the prompt token immediately preceding the first response token. Before joint adaptation, a matched-capacity depth study fits identical logistic probes at three evenly spaced API indices, 9, 18, and 27, using source-group-disjoint training-side diagnostic partitions; the development comparison selects index 27. Because index zero is the embedding output, this readout is the output of transformer block 26. Benchmark test records and external probes never enter this choice.

Let $\mathbf{h}_{27}$ denote the selected representation and $\mathbf{z} = \text{Norm}(\mathbf{h}_{27})$, where Stage 1 normalization statistics are computed from frozen-backbone training representations. The auxiliary state gate is $\mathbf{p}_\phi = \text{softmax}(g_\phi(\mathbf{z}))$, $\mathbf{p}_\phi = (p_{\phi,A}, p_{\phi,C}, p_{\phi,U})$, where A, C, and U denote Supported, Visual Contradiction, and Target Unavailable. The gate g_ϕ is a $4096 \to 512 \to 256 \to 3$ MLP with GELU activations and dropout 0.1.

We adapt the backbone with LoRA (Hu et al. 2022) by

$$\mathcal{L}_t = \mathcal{L}_{\text{LM}} - \lambda_t N^{-1} \sum_i w_{s_i} \log p_{\phi,s_i} \quad (5)$$

where $\lambda_t = 0.4 \min(1, t/10)$, $\mathbf{w} = (1.0, 1.5, 1.0)$, t is the optimizer step, $\mathcal{L}_{\text{LM}}$ is cross-entropy over the structured grounding or abstention response, N is the minibatch size, and $\mathbf{w}$ orders the three states as (A, C, U). Same-source

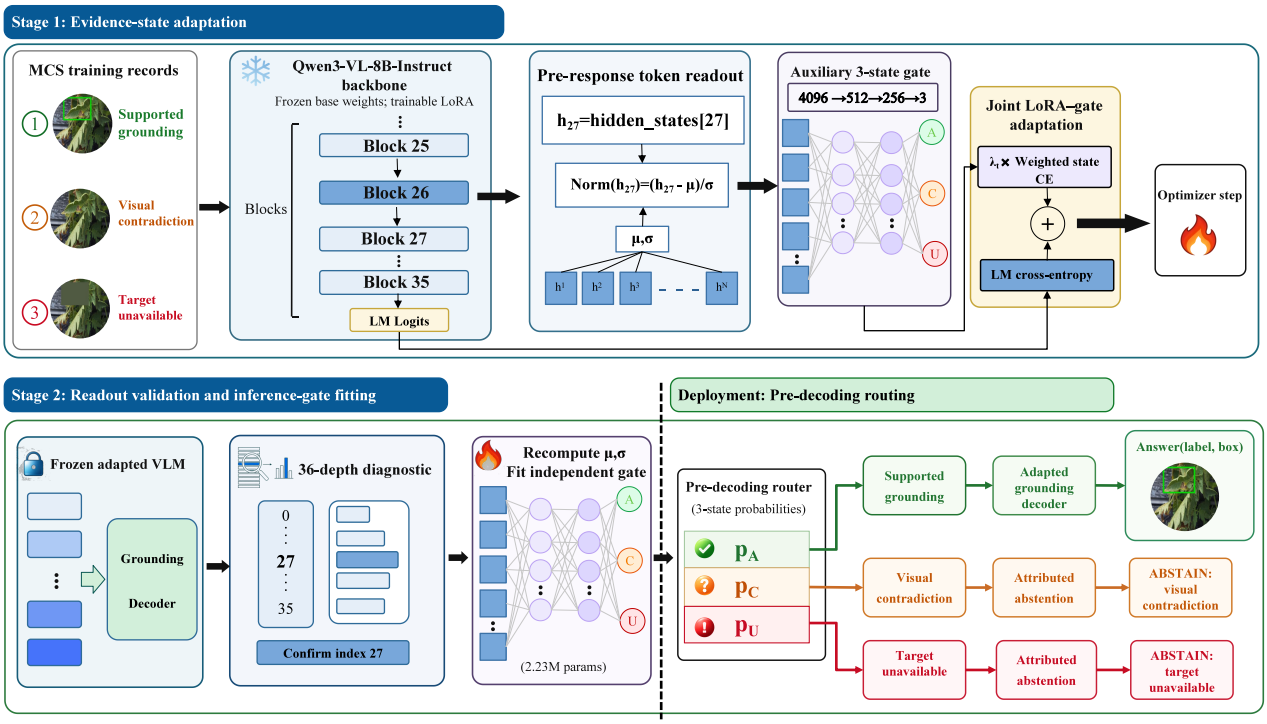


Figure 2: Two-stage ReGate. Stage 1 jointly adapts LoRA and an auxiliary state gate; Stage 2 freezes the adapted VLM, confirms the layer-27 readout, and fits the parameterized inference gate for supported grounding or reason-attributed abstention.

counterfactual records encode the metamorphic structure directly in their evidence-state labels, enabling joint representation adaptation through standard sample-wise optimization without an additional pairwise objective. Among the six epoch checkpoints, we retain the checkpoint with minimum teacher-forced language-model loss on a fixed 100-record subset of the formal 2,408-record development pool.

4.2 Readout Validation and Inference-Gate Refit

After adaptation, a separate source-group-disjoint diagnostic fits a logistic probe P_ℓ at each of the 36 transformer depths using 1,599 probe-fitting and 137 probe-development records. Over states $\mathcal{S} = \{A, C, U\}$, it selects the layer with maximum worst-state validation accuracy:

$$m_\ell = \min_{c \in \mathcal{S}} \mathrm{Acc}_{\mathrm{val},c}(P_\ell), \qquad \ell^* = \arg\max_\ell m_\ell. \tag{6}$$

The scan independently confirms index 27 with $m_{27} = 0.9574$. These probes use only training-side diagnostic records and do not enter gate fitting or inference.

We then freeze the selected backbone and grounding decoder, recompute normalization statistics from all adapted training representations, and fit an independently parameterized inference gate g_ψ with the same MLP architecture using cross-entropy and development-set early stopping. The Stage 1 gate serves only as an auxiliary training head; Stage 2 does not reuse its parameters, and only g_ψ supplies reported ReGate decisions. This post-adaptation fit separates representation learning from final routing and estimates the deployed decision rule on the stationary adapted feature space. This stage reuses the same evidence-state labels and introduces no additional supervision. Including biases, the inference gate has $2.23M$ parameters, or 0.028% of the backbone.

4.3 Reason-Attributed Routing

The final gate produces $(p_A, p_C, p_U) = \mathrm{softmax}(g_\psi(\mathrm{Norm}(\mathbf{h}_{27})))$. The default operating point predicts $\hat{s} = \arg\max_{s \in \{A,C,U\}} p_s$, with ties involving A resolved in favor of answering. A supported decision invokes the adapted grounding decoder with the shared structured instruction to generate a label and box; C and U directly return visual contradiction and target unavailability, respectively, without autoregressive grounding generation. For selective operation, thresholded routing answers when $p_A \geq \tau$ and otherwise chooses $\arg\max_{s \in \{C,U\}} p_s$. The reported low-FCA operating point selects τ exclusively on development data, while sweeping τ traces the FCA–retention curve.

5 Experiments

We evaluate whether evidence-state adaptation reduces unsupported grounding, same-source construction produces compatibility-sensitive representations, and explicit routing transfers across datasets and decision interfaces.

5.1 Experimental Setup

Systems. Base is the frozen Qwen3-VL backbone with native structured decoding, **SFT native** applies the same decoder after Stage 1 adaptation, and **ReGate** adds the Stage 2 gate. Base–SFT evaluates the complete adaptation pipeline:

Dataset	Interface	FCA↓	Ret.↑	S3↑	E2E↑
Comprehensive	Base	0.211	0.833	0.660	0.656
	SFT native	**0.094**	0.851	0.819	**0.705**
	ReGate	0.132	**0.976**	**0.856**	0.701
PlantWild	Base	0.360	0.792	0.375	–
	SFT native	**0.155**	0.764	0.647	–
	ReGate	0.269	**0.854**	**0.694**	–
RefCOCOg	Base	**0.044**	0.983	0.762	0.525
	SFT native	**0.044**	0.901	**0.834**	**0.645**
	ReGate	0.083	**1.000**	0.820	**0.645**

Table 3: Base–SFT–ReGate comparison at fixed operating points. Ret., S3, and E2E denote supported retention, Strict-3, and E2E-mIoU; PlantWild has no box annotations.

Probe/interface	MTS↑	MCC↑	FCA↓	A_FU↓
Comp.-Gate	.820/.527	.849/.523	.132/.486	.024/.045
Comp.-Native	.807/.756	.758/.660	.094/.193	.149/.148
Ref.-Gate	.812/.867	.920/.940	.083/.055	.000/.006
Ref.-Native	.826/.876	.858/.923	.044/.039	.099/.039
PW-Gate	.673/.619	.560/.350	.269/.250	.146/.396
PW-Native	.648/.608	.602/.449	.155/.322	.236/.208
L1-Gate	.651/.502	.393/.085	.615/.766	.059/.166
L2-Gate	.622/.410	.570/.091	.476/.878	.018/.068
L3-Gate	.804/.502	.836/.318	.176/.831	.011/.000

Table 4: Matched construction contrast. Cells report Grouped/Independent. Comp., Ref., and PW denote Comprehensive, RefCOCOg, and PlantWild.

evidence-state data, joint language-model and state losses, and LoRA. SFT–ReGate measures the effect of adding the deployed Stage 2 router while keeping the adapted backbone and grounding generator fixed. LoRA uses $r = 16$, $\alpha = 32$, dropout 0.05, learning rate 10^{-4}, and at most six epochs.

Routing interfaces. Five generative interfaces use the same adapted backbone and generator. **Native LM** is SFT native with the base structured instruction; **Direct** enumerates the three decisions; **Reward** adds outcome-utility guidance; **R+E** further specifies evidence-preservation and evidence-change principles; and **S-CoT** requests staged support analysis before the same output schema. Across these interfaces, prompt examples, preprocessing, decoding, schema, parser, and evaluation denominators are fixed; only the routing instruction changes. ReGate instead obtains the decision from its Stage 2 gate. GPT-5.6 Sol serves as a prompted general-purpose reference under the same task schema. All numerical comparisons use the common three-state output space. GREC and RECANTFormer (He et al. 2023; Hemanthage et al. 2024) return an empty target set, while Counterfactual REC uses a single counterfactual label; their native outputs therefore do not distinguish visual contradiction from target unavailability.

Evaluation protocol. We use the six probes in Table 2 and the metrics in Section 3.3. The Grouped/Independent comparison retrains the complete pipeline with matched backbone, objective, state counts, and optimization. Additional details, including prompt strings, computing environment, source-separation procedures, full FCA–retention curves, and uncertainty estimates are provided in the supplement.

5.2 Main Results

Evidence-state adaptation. Stage 1 lowers FCA from 0.211 to 0.094 on the comprehensive test and from 0.360 to 0.155 on PlantWild, while preserving 0.044 on RefCOCOg. It simultaneously improves E2E-mIoU from 0.656 to 0.705 in-domain and from 0.525 to 0.645 on image-disjoint RefCOCOg (Table 3), enhancing label-gated localization while transferring hallucination mitigation to natural field imagery.

High-retention routing. With the adapted backbone and generator fixed, the argmax gate raises comprehensive-test retention from 0.851 to 0.976, MCC from 0.758 to 0.849, and Strict-3 from 0.819 to 0.856, while reducing A_FU from 0.149 to 0.024. This argmax operating point prioritizes supported coverage, reaching 0.976 retention with FCA 0.132; the development-selected threshold below provides the lower-FCA operating point.

Development-selected operation. Selecting $\tau = 0.9888$ exclusively on all $2{,}408$ development records at matched 86.05% development retention yields 88.96% test retention and 4.54% FCA, compared with 85.11% retention and 9.45% FCA for SFT native. Development-selected routing therefore increases supported coverage while reducing false commitment by 52.0%. All fixed ReGate entries use three-class argmax; this thresholded point is reported separately.

5.3 Construction and Evidence Analyses

Same-source construction. We compare Grouped training with independently sampled, category-constrained negatives through matched full retraining. The two conditions preserve supported records, state counts, targets, category–state strata, objective, architecture, and hyperparameters, and each fits its own final gate.

Under the matched seed-42 protocol, Grouped training raises comprehensive-test Strict-3 from 0.663 to 0.856, retention from 0.955 to 0.976, supported answer-and-grounding accuracy from 0.762 to 0.794, unsupported-reason accuracy from 0.504 to 0.865, and the T6 pass rate from 0.263 to 0.571. It also reduces PlantWild A_FU from 0.396 to 0.146 and raises retention from 0.604 to 0.854. Independent negatives remain competitive on RefCOCOg, whereas Grouped training produces the largest gains on the compatibility-controlled and source-disjoint probes (Table 4), validating same-source construction for reason-attributed decisions.

Representation readout. Using identical source-group-disjoint train/development/test partitions and the same classifier over five probe-fitting seeds with the base backbone frozen, the pre-response prompt token reaches MCC 0.769 ± 0.020, compared with 0.363 ± 0.011 for visual-token mean, 0.337 ± 0.009 for visual-token max, and 0.115 ± 0.029 for visual query-attention. It is the strongest tested readout; the separate post-adaptation depth scan selects index 27.

Controlled evidence response. The probes isolate fine-grained same-crop contradiction (L1 FCA 0.615), complete evidence removal (L2 FCA 0.476), and graded visibility

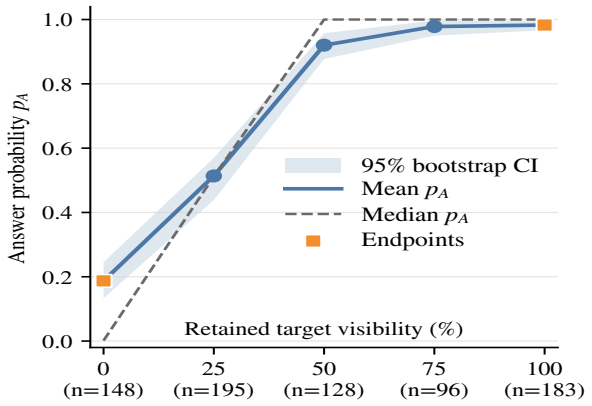


Figure 3: L3 visibility response. Mean and median p_A increase with retained target visibility; shading shows the 95% source-instance-group bootstrap CI, and orange markers denote the hard-evaluation endpoints.

Interface	Comprehensive	RefCOCOg	PlantWild
ReGate	**.849/.024/.856**	**.920/.000**/.820	.560/**.146**/.694
Native LM	.758/.149/.819	.858/.099/**.834**	**.602**/.236/.647
Direct	.724/.198/.844	.862/.127/.693	.581/.250/**.755**
Reward	.703/.162/.834	.838/.127/.588	.506/.208/.645
R+E	.696/.180/.832	.768/.204/.627	.513/.222/.679
S-CoT	.637/.357/.765	.685/.343/.572	.545/.417/.642

Table 5: Routing interfaces results on MCC/A_FU/Strict-3.

(L3 MCC/FCA $0.836/0.176$). Among the balanced $395/395$ L2 records, source-instance and target-label matching after deduplication yields 148 complete target-retaining/target-removal pairs. Removing the target lowers p_A by 0.786 on average (95% CI $[0.720, 0.840]$), with the expected direction in 96.6% of pairs. Across 183 L3 groups, p_A increases with retained visibility (Spearman $\rho = 0.750$, Kendall $\tau = 0.682$); 78.6% of adjacent levels are correctly ordered, corresponding to a 21.4% violation rate (Fig. 3). The L2 fixed-point FCA complements this probability-level analysis by testing whether the response crosses the final decision boundary.

Relation-level consistency. The strict-pass vector is T1 paraphrase 0.818; T2 photometric/crop-out $0.798/0.938$; T3 geometry/crop-out $0.808/0.913$; T4 control 0.831; T5 absent query 0.948; T6 confusable query 0.571; T7 inclusion/ladder $0.750/0.781$; and base crop-out 0.859, yielding macro MTS 0.820. Joint success on the supported T4 control and unavailable crop-outs confirms state-specific behavior across transformations. The most demanding relation, T6, demonstrates discrimination among fine-grained same-crop alternatives.

5.4 Routing Interfaces and Generalization

Routing-interface comparison.

ReGate leads six of nine dataset–metric comparisons: all three comprehensive metrics, RefCOCOg MCC and A_FU, and PlantWild A_FU (Table 5). Native LM leads RefCOCOg Strict-3 and PlantWild MCC, while Direct leads PlantWild Strict-3. Overall, explicit evidence gating provides the strongest cross-dataset routing profile.

Probe	FCA↓	A_FU↓	MCC↑	S3↑	E2E↑
Comp.	.132/.302	.024/.076	.849/.639	.856/.664	.701/.625
L1	.615/.576	.059/.102	.393/.365	.651/.649	.810/.614
L2	.476/.570	.018/.073	.570/.411	.682/.656	.641/.573
L3	.176/.338	.011/.077	.836/.615	.807/.785	.689/.657
Ref.	.083/.088	.000/.000	.920/.915	.820/.677	.645/.446
PW	.269/.223	.146/.167	.560/.587	.694/.463	–/–

Table 6: ReGate/GPT-5.6 Sol comparison; cells follow this order. S3 and E2E denote Strict-3 and E2E-mIoU.

Source-disjoint transfer. On source-disjoint PlantWild, ReGate reaches state-specific strict accuracies of $0.750/0.317/0.951$ for supported grounding, visual contradiction, and target unavailability. The strong supported and unavailable results confirm transfer to natural field imagery, while the contradiction score establishes fine-grained same-crop discrimination as the benchmark's most demanding transfer test.

General-purpose reference. Table 6 compares the complete two-stage ReGate system with prompted GPT-5.6 Sol on all six probes. The comparison uses identical records, task schema, parser, and ground-truth denominators; GPT-5.6 Sol receives the Native-LM instruction and per-record query of ReGate's supported route. The same fixed instruction is retained on RefCOCOg without dataset-specific prompt tuning.

On the comprehensive test, ReGate/GPT-5.6 Sol/SFT native attain A_FU $0.024/0.076/0.149$ and MCC $0.849/0.639/0.758$; ReGate and SFT native also exceed the prompted reference in E2E-mIoU. Paired 95% intervals favor ReGate over GPT-5.6 Sol for comprehensive-test FCA, A_FU, Strict-3, and E2E-mIoU. The L1 and PlantWild FCA intervals overlap, while paired intervals favor ReGate for L1 E2E-mIoU, L2/L3 FCA, and RefCOCOg Strict-3 and E2E-mIoU. These results establish strong transfer under a shared-prompt protocol and a consistent advantage on the comprehensive evaluation.

6 Conclusion

We introduced reason-attributed selective grounding, metamorphic counterfactual supervision (MCS), and the two-stage ReGate framework. MCS converts deterministic same-source relations into annotation-defined supervision for supported grounding, visual contradiction, and target unavailability. ReGate combines representation adaptation with compact pre-decoding routing: Stage 1 reduces unsupported commitment and improves label-gated localization, while Stage 2 enables high-retention routing and a development-selected operating point that halves FCA versus SFT native while increasing retention. Matched construction studies, controlled evidence interventions, source-disjoint transfer, and a shared-prompt GPT-5.6 comparison demonstrate compatibility-sensitive, evidence-responsive decisions. By reusing existing localization annotations, MCS provides a scalable auxiliary task for VLM pretraining, while ReGate offers a backbone-compatible foundation for extending selective grounding across models and visual domains.

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